

Jeffrey Xu

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EDUCATION

University of Waterloo

Bachelor of Applied Science, Honours, Electrical Engineering

- 3.72 GPA

Waterloo, ON

Sep 2025 – Apr 2030

SKILLS

Hardware & Lab: Altium Designer, FPGA, Analog/Digital Circuits, Circuit Analysis, Component Selection, Oscilloscope, Digital Multimeter (DMM), Soldering, Arduino, Raspberry Pi

Software & Firmware: C, C++, Python, VS Code, GitHub, Verilog, Quartus Prime, SPI, I2C

EXPERIENCE

Electrical Hardware Designer

Sep 2025 – Present

UW Orbital

Waterloo, ON

- Designed an EPS current-sense circuit using an INA180 amplifier and a shunt resistor, selecting an appropriate shunt value to limit power loss.
- Followed INA180 datasheet layout guidance and electrical standards by checking capacitor voltage rating, and shunt power rating to improve measurement stability and reduce noise in the current-sense path.

Research & Quality Assurance Engineering Intern

Jan 2026 – Apr 2026

Groupthink Labs

Waterloo, ON

- Identified 30+ target accounts and managed outreach for pilot user recruitment using Excel.
- Conducted stakeholder interviews with design leads to establish key software QA priorities.
- Created 6 comprehensive mock pilots mapping edge cases to expand QA validation coverage.

PROJECTS

Digital HVAC & Energy Controller | *Verilog, FPGA, Quartus Prime*

- Designed a Verilog HVAC controller using state machines for temperature feedback loops.
- Mapped physical FPGA push-buttons to simulate door, window, and vacation mode inputs.
- Simulated control logic using Quartus Prime waveform editors to verify expected outputs.

Analog Circuit Modelling & Verification | *Circuit Analysis, Lab Instruments*

- Modelled analog circuits on breadboards, verifying experimental data against theoretical models.
- Analyzed electrical signals using oscilloscopes to validate function generator inputs.
- Tested circuit stability under shifting component loads using decade resistance boxes.

Autonomous Fleets Challenge | *Python, Raspberry Pi*

- Competed in the Toyota 2026 Challenge for autonomous robotic fleet warehouse coordination.
- Programmed a PRIZM robot using computer vision for AprilTag tracking and line following.
- Tested physical navigation over a Tailscale VPN, tuning Python path-correction scripts.

Automated Blinds Control System | *Arduino, C, SPI*

- Developed an automated blinds framework using C to control stepper motor state profiles via SPI commands.
- Programmed the microcontroller's internal Analog-to-Digital Converter (ADC) to sample ambient light levels from a photoresistor sensor voltage divider every 10 minutes.
- Implemented power-saving idle states between polling intervals to optimize system energy efficiency.

Portable Sensory Cube | *Arduino, SPI, I2C*

- Built an Arduino-based medical assistive device utilizing SPI to drive an LCD character display aimed at mitigating hospital induced delirium.
- Integrated a MAX30102 pulse oximeter sensor via I2C, configuring register settings to sample real-time photoplethysmography (PPG) data for heart rate processing.